Descriptive Analytics Report: Goodreads

Best 1500 Books — Reader Ratings & Publication Behavior

Name: Kumar Ashutosh

Roll no: 1250258249

Course: BCA (DS& AI)

Section: BCADS24

Date: 15th August, 2026

Tracking Publication Trends, Genre Categories, Rating Performance, Author Contribution & Reader Engagement

Note on data: This report uses the supplied Goodreads dataset containing 1,500 book records spanning 49 genres, 873 authors and publication years 2001–2020. Of these, 26 records had corrupted rating fields (non-numeric text in place of numbers) and were excluded from rating-based calculations; all 1,500 records are retained for count-based and genre-based analysis. The dataset is a curated "best books" list rather than a full catalogue of published books. Therefore, the analysis describes the supplied observations and does not assume a larger real-time Goodreads catalogue.

1. Objective

Build a descriptive analytics layer over Goodreads book data to answer the following recurring questions:

- How many books were published per year in this list?
- How is the average rating trending across publication years?
- Which genre contributes the highest number of books?
- Which genres contribute the most reader engagement (raters)?
- Which genres have the highest average ratings?
- Overall rating performance across the dataset.
- Which authors contribute the most books to this list?
- How does reader engagement (raters) relate to Goodreads score?
- Publication-era rating patterns (2001–2010 vs 2011–2020).
- Distribution of ratings across all books.

---

2. How This Would Work at Real Scale

Processing a full-scale book-rating catalogue is an engineering problem before it is an analytics problem. A realistic architecture would collect book and rating events, aggregate them by genre/author/year, store raw and aggregated data, and expose the resulting KPIs through a data visualization layer.

Layer	Purpose	Typical tools
Investigation	Research, understand and prepare the book dataset	Google, Claude & Chat GPT
Stream processing	Aggregate by genre, author and year	JUPYTER notebook (python)
Storage	Retain raw and processed data	Excel files/CSV files
Batch aggregation	Rating KPIs and summary metrics	JUPYTER Notebook
Report	Present analysis, graphs and insights	Data Visualization/MS Word

3. Books Published Per Year

Across the supplied **1,500-book list**, records span publication years **2001 to 2020**. The chart below shows how the number of books in the list varies by publication year.

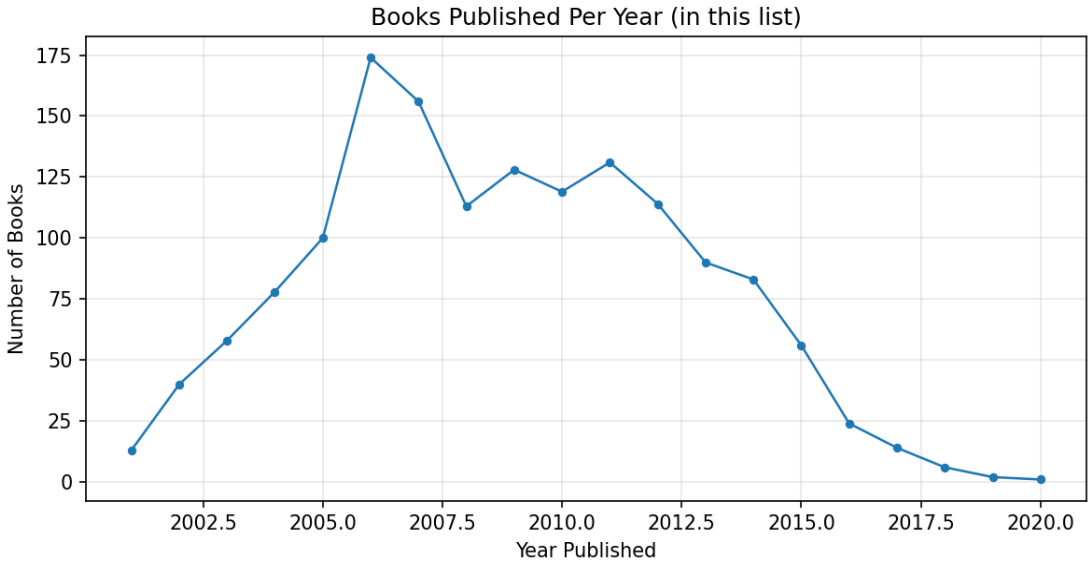


Figure 1. Books published per year

INSIGHTS:

- Years with more listed books indicate stronger representation in this curated list, not necessarily higher real-world publishing output.
- Years with fewer listed books indicate comparatively weaker representation.
- Monitoring per-year counts can help identify which publication periods this list favors.
- This can support understanding of the list's coverage and potential selection bias.

KPI :

Total books: **1,500**

Peak year: **2006 (174 books)**

Year range: **2001 – 2020**

4. Average Rating Trending Over Publication Years

The overall average rating across the supplied records is **~4.01 out of 5**. The average rating per publication year is **calculated directly from the dataset's avg_rating field** (26 corrupted rows excluded).

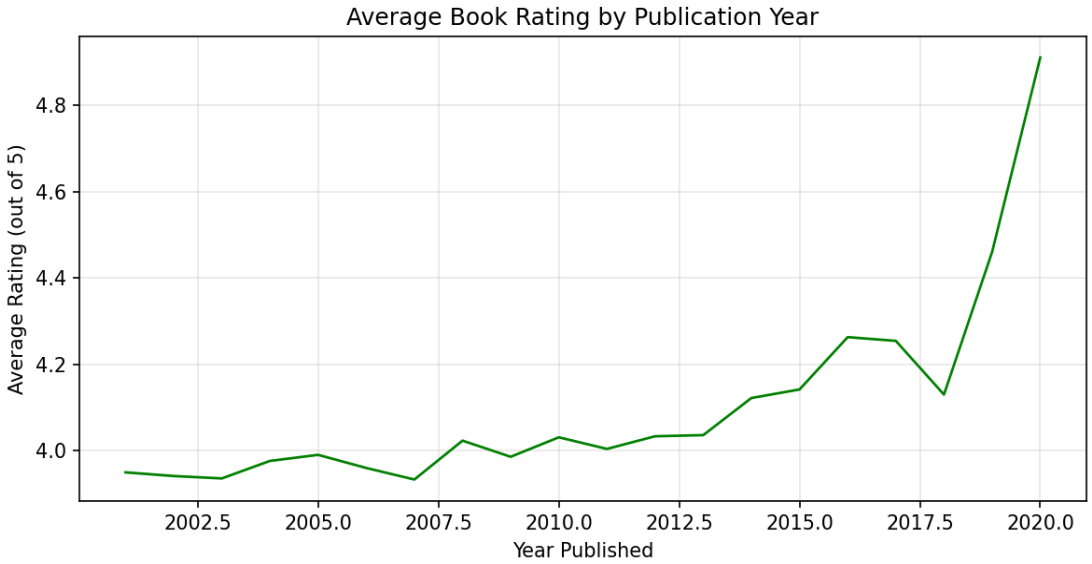


Figure 2. Average rating by publication year

INSIGHTS:

- Average rating varies modestly across the available years.
- Higher-rating years indicate publication periods with stronger-received titles in this list.
- The most recent years (2018–2020) show elevated averages, but represent very few books (6, 2 and 1 book respectively) and should be read with caution.
- Rating trend monitoring helps identify which eras contributed the best-received titles to this list.

KPI :

Overall average rating: **4.01 / 5**

Median rating: **4.02 / 5**

Highest-rated year (n≥20 books): **2016 (avg 4.26)**

5. Best-Performing Genres by Book Count

The dataset spans **49 distinct genres**. **Fiction** is the largest single genre, accounting for **385 books (25.67%)** of the list, followed by Fantasy and Young Adult.

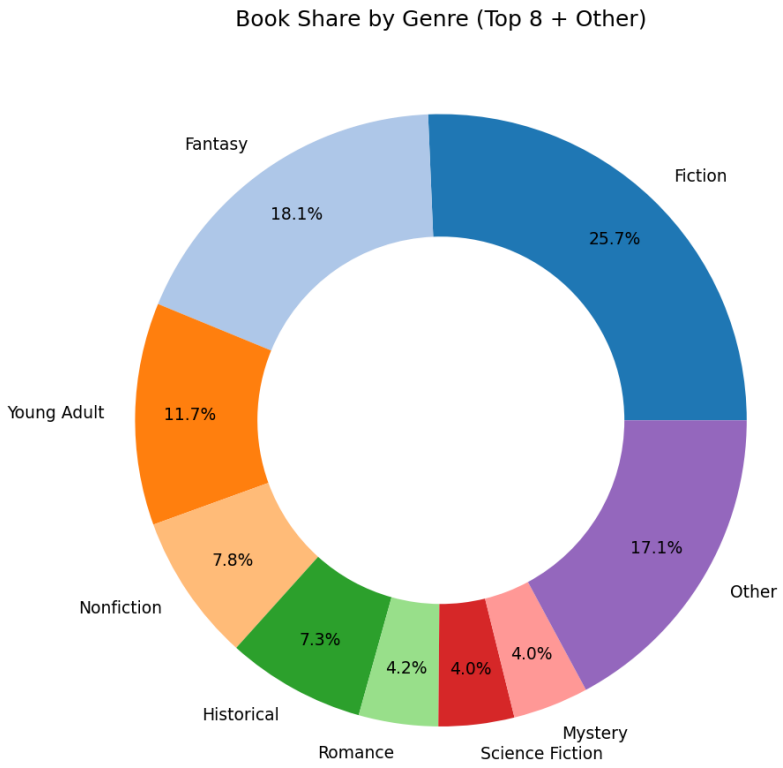


Figure 3. Book share by genre (top 8 + other)

INSIGHTS:

- Genre representation is concentrated: the top 3 genres (Fiction, Fantasy, Young Adult) together account for over half the list.
- No conclusion should be drawn that Fiction is the most popular genre in the real world — this reflects only the supplied curated list.
- The long tail of 40+ smaller genres shows the list still covers a wide variety of niche categories.

KPI :

Top genre by count: **Fiction**

Book share: **≈25.67%**

Genres represented: **49**

6. Genres Contributing the Most Reader Engagement

Reader engagement, measured as total number of raters, is led by **Young Adult** with **~58.5 million total raters**, narrowly ahead of Fiction (~55.9 million) and Fantasy (~47.9 million).

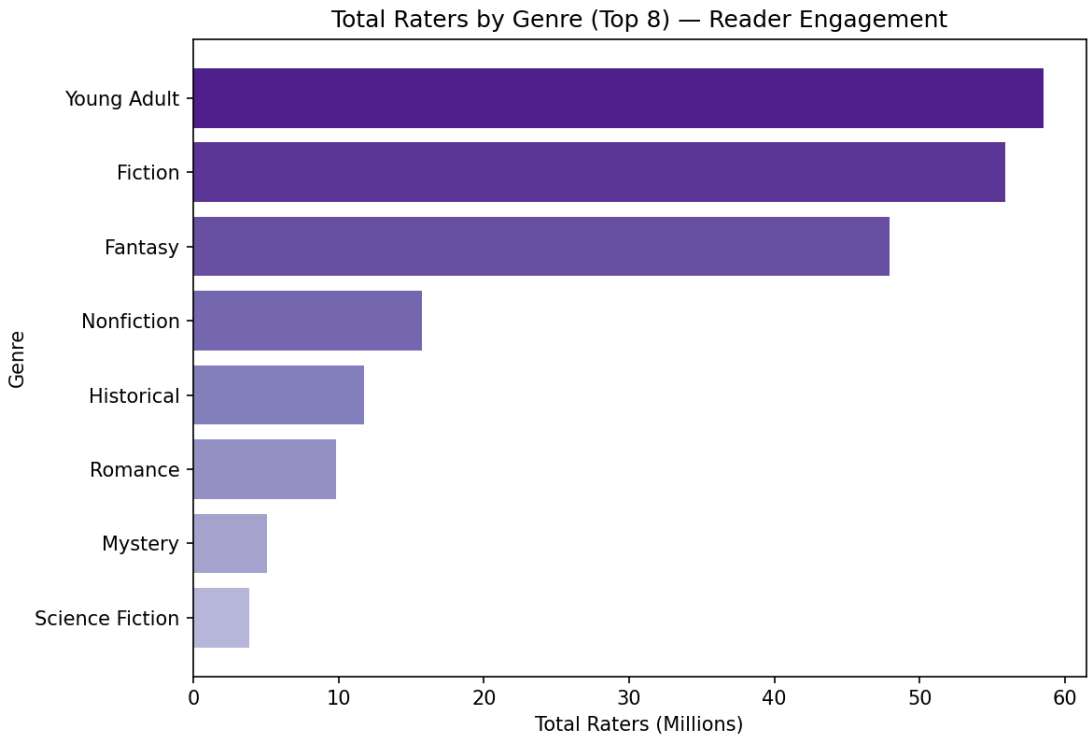


Figure 4. Total raters by genre (top 8)

INSIGHTS:

- Engagement ranking differs from the book-count ranking — Young Adult leads engagement despite having fewer books than Fiction.
- This suggests Young Adult titles in this list attract disproportionately large reader bases per book.
- A genre with fewer books but very high engagement may represent especially popular series (e.g., Harry Potter, The Hunger Games).
- Engagement should be considered alongside book count to understand genre performance more completely.

KPI :

Top genre by raters: **Young Adult**

Total raters (Young Adult): **≈58.53 Million**

Total raters across dataset: **≈222.6 Million**

7. Genres With the Highest Average Ratings

Among genres with at least 20 books in the list, **Fantasy** has the highest average rating at **~4.15**, followed by Thriller (~4.12) and Romance (~4.11).

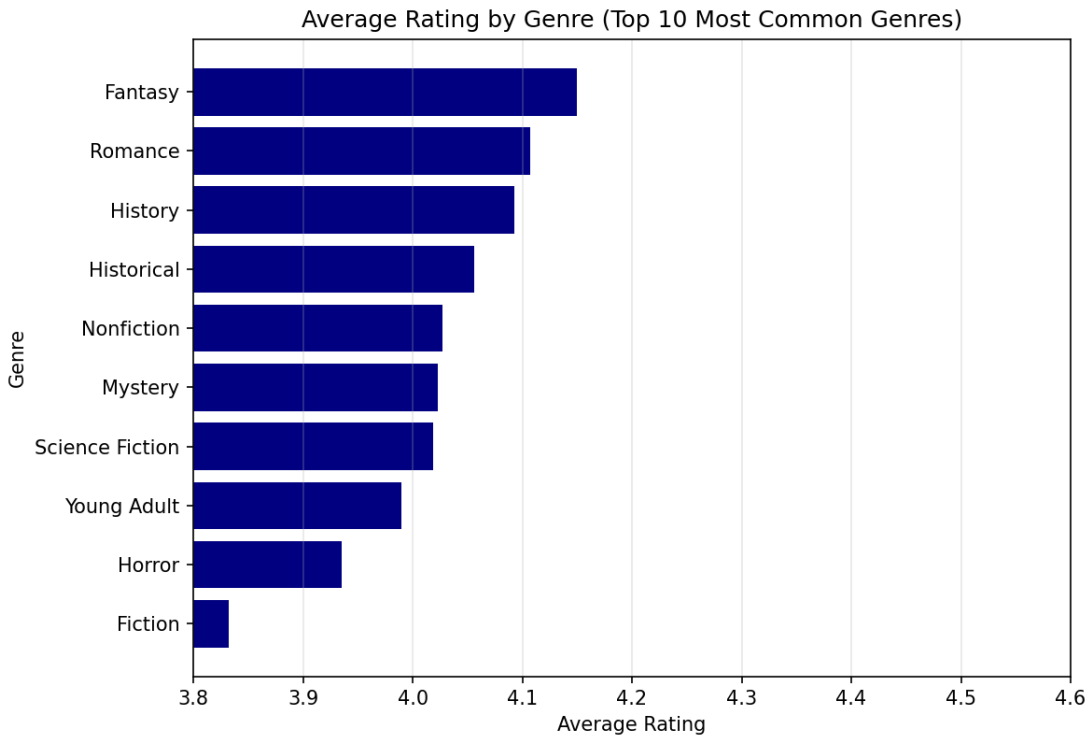


Figure 5. Average rating by genre (top 10 most common genres)

INSIGHTS:

- Fantasy combines both strong book-count representation and the highest average rating among well-represented genres.
- Genres with fewer than 20 books were excluded from this ranking to avoid small-sample distortion.
- Young Adult, despite leading in engagement, has a comparatively lower average rating (~3.99) among the top genres shown.
- Rating differences across genres are moderate — no genre in the top 10 falls below a 3.9 average.

KPI :

Highest-rated genre (n≥20): **Fantasy (4.15)**

Lowest of top 10 (n≥20): **Young Adult (3.99)**

8. Total Books by Genre

Fiction is also the largest contributor when ranked strictly by book count, at **385 books**. The distribution shows that genre representation is concentrated in a subset of categories rather than evenly spread across all 49 genres.

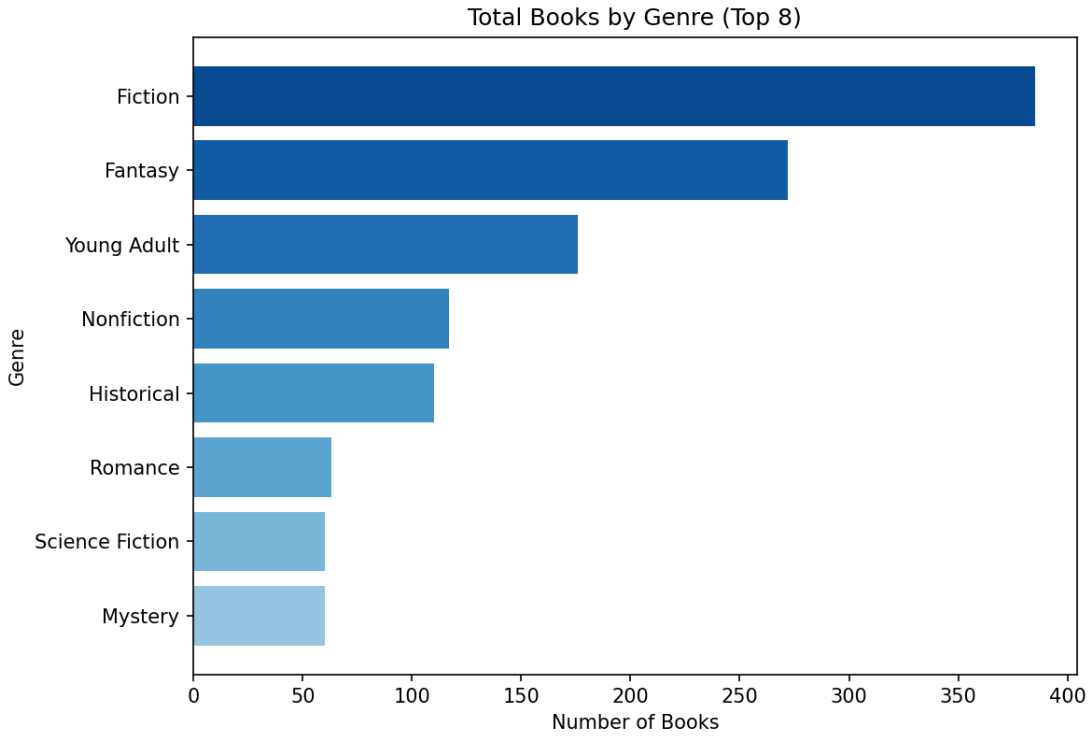


Figure 6. Total books by genre (top 8)

INSIGHTS:

- This confirms concentration: the top 8 genres account for the large majority of the 1,500-book list.
- Genre concentration is common in curated "best of" lists, which tend to favor commercially dominant categories.
- Publishers or platforms curating similar lists may want broader genre coverage for diversity.

KPI :

Top-performing genre: **Fiction**

Recorded books: **385**

Share of total list: **≈25.67%**

9. Average Rating Performance

The average rating across the supplied records is **4.01 / 5**.

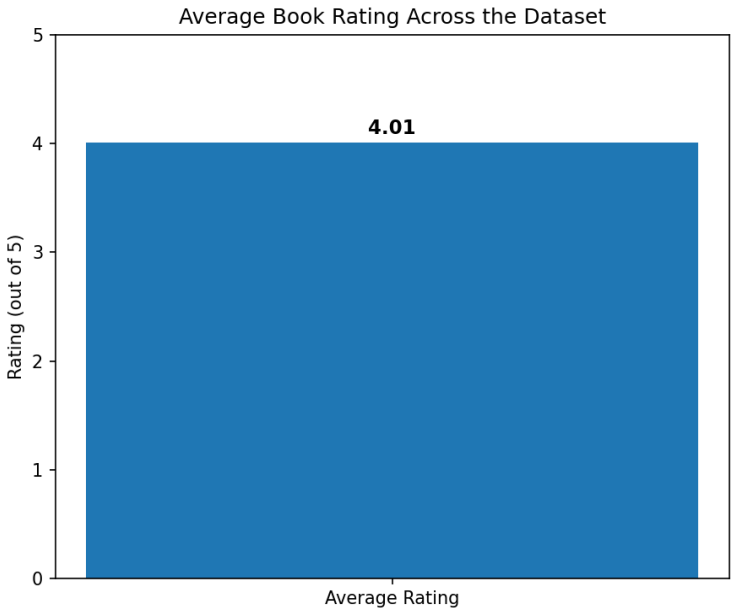


Figure 7. Average book rating across the dataset

INSIGHTS:

- Ratings across this curated list skew positive, as expected for a "best books" collection rather than a random sample.
- The median rating (4.02) is close to the mean, indicating a fairly symmetric rating distribution.
- Rating performance should be monitored alongside reader engagement (raters), since a high rating from very few raters carries less weight than one from millions.

KPI :

Average rating: **4.01 / 5**

Median rating: **4.02 / 5**

Rating data availability: **1,474 / 1,500 books (98.3%)**

10. Top-Contributing Authors

Reader demand for individual authors is distributed across **873 unique authors** in the list. **Rick Riordan** and **Stephen King** tie for the most books represented, each with **16 titles**.

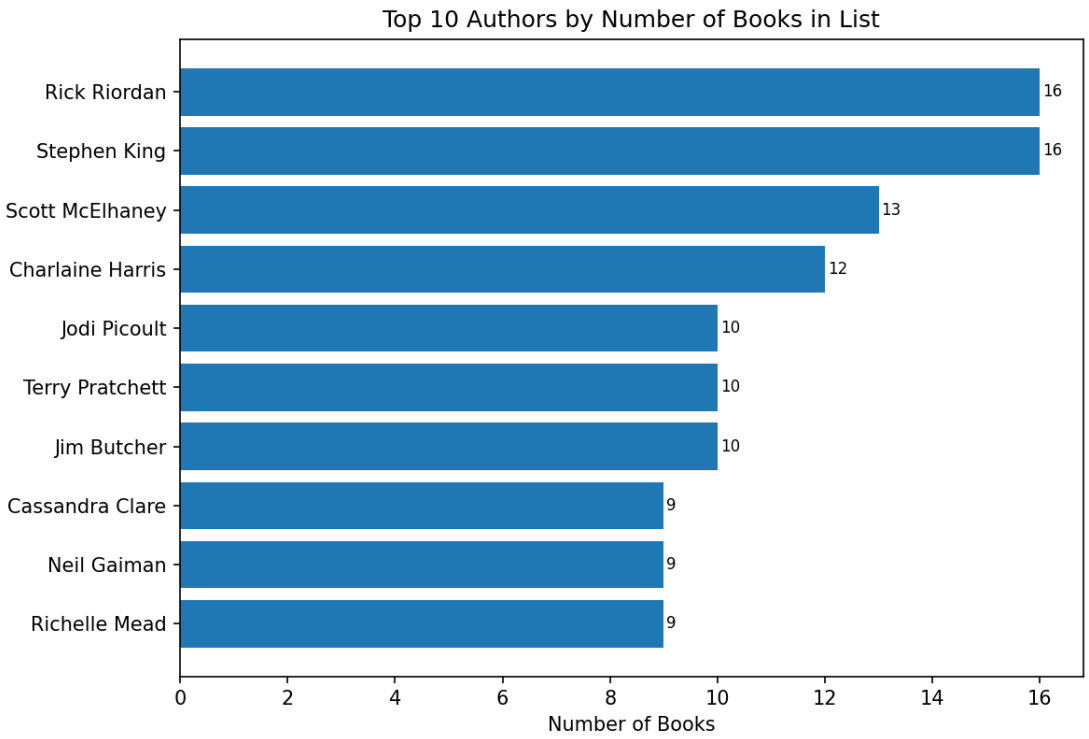


Figure 8. Top 10 authors by number of books in list

INSIGHTS:

- Author contribution is concentrated: the top 10 authors together account for over 110 of the 1,500 books, despite representing just over 1% of all authors.
- Prolific series authors (Riordan, King, Harris, Butcher) tend to dominate multi-book representation.
- Author-level analysis can support understanding which writers this list systematically favors.
- This concentration is expected in a genre-heavy list weighted toward Fantasy and Young Adult series.

KPI :

Top-performing authors: **Rick Riordan & Stephen King**

Books each: **16**

Total unique authors: **873**

11. Reader Engagement vs Goodreads Score

The relationship between number of raters and Goodreads score provides an additional descriptive view of book-level popularity. Books with more raters tend to have higher scores, but the scatter plot should be interpreted as **association rather than proof of causation**.

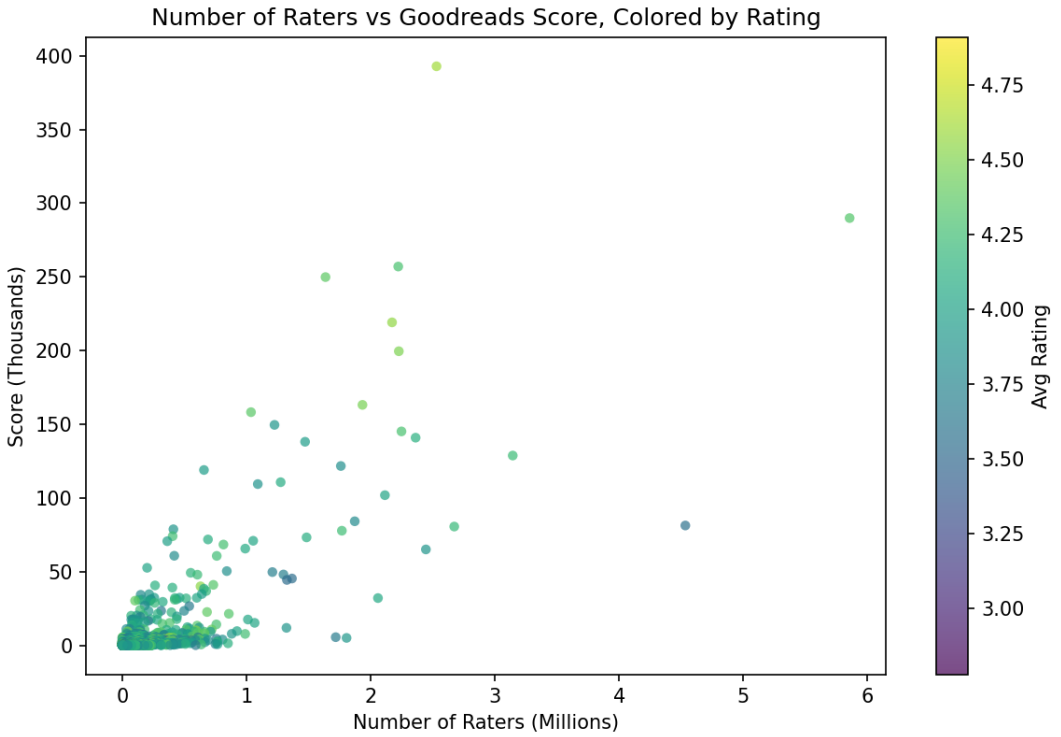


Figure 9. Number of raters vs Goodreads score, colored by average rating

INSIGHTS:

- A positive association is visible: books with millions of raters generally also carry very high Goodreads scores.
- However, the graph does not prove that more raters directly cause a higher score — score itself is partly derived from voting volume and ranking behavior.
- Color (average rating) does not show a strong independent pattern here — high scores appear across a range of rating values.
- This analysis can support understanding of which books are both widely read and highly regarded.

KPI :

Most-rated book: **The Hunger Games (5.86M raters)**

Highest-score book: **Harry Potter and the Deathly Hallows (score 392,793)**

Relationship: **Higher rater counts are associated with higher scores in the supplied data**

12. Publication-Era Rating Pattern (2001–2010 vs 2011–2020)

Across the row-level observations, average ratings are **3.98** for books published **2001–2010** (979 books) versus **4.07** for books published **2011–2020** (521 books).

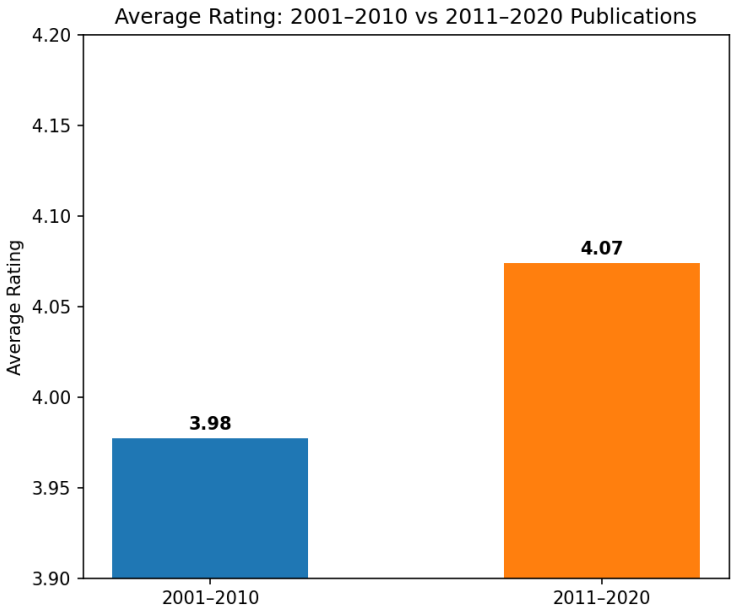


Figure 10. Average rating: 2001–2010 vs 2011–2020

INSIGHTS:

- More recently published books in this list carry a modestly higher average rating than earlier ones.
- This could reflect changing reader rating behavior, tighter curation of recent bestsellers, or fewer but more selectively chosen recent titles (521 vs 979 books).
- The dataset is a curated list rather than a complete publishing record, so this should be treated as a descriptive comparison of the available observations, not a general trend in publishing.

KPI :

2001–2010 average rating: **3.98 (979 books)**

2011–2020 average rating: **4.07 (521 books)**

13. Distribution of Ratings

The overall rating distribution shows how individual book ratings are spread across the dataset, useful for understanding the shape of quality perception beyond a single average figure.

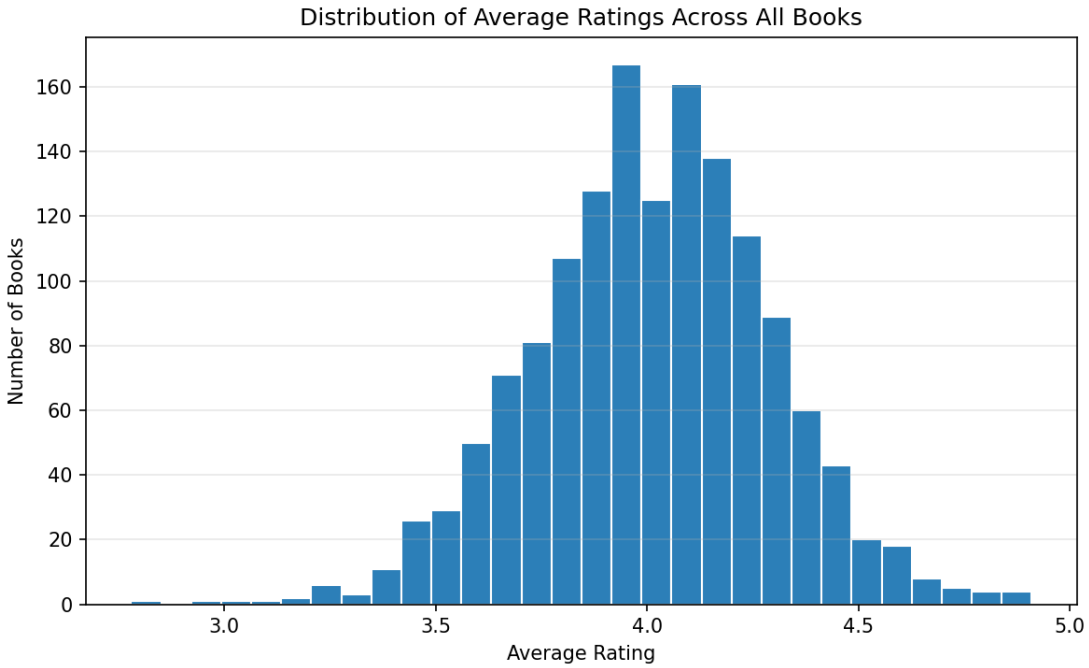


Figure 11. Distribution of average ratings across all books

INSIGHTS:

- The distribution is concentrated between roughly 3.7 and 4.3, consistent with a curated "best books" list rather than a full random sample of all published books.
- Very few books fall below 3.5 or above 4.7, indicating the list is pre-filtered toward generally well-received titles.
- Understanding distribution shape (not just the mean) helps identify how tightly clustered reader opinion is across this list.

KPI :

Rating range (bulk of data): **≈3.7 – 4.3**

Books with data: **1,474**

14. Top 10 Highest-Rated Books (Min. 10,000 Raters)

To avoid small-sample distortion, this ranking is restricted to books with at least 10,000 raters. The table below lists the ten highest-rated qualifying books in the supplied dataset.

Book	Author	Genre	Rating	Raters
The Complete Calvin and Hobbes	Bill Watterson	Sequential Art	4.82	33,280
Words of Radiance (Stormlight Archive #2)	Brandon Sanderson	Fantasy	4.76	171,883
A Court of Mist and Fury (ACOTAR #2)	Sarah J. Maas	Fantasy	4.67	248,558
The Way of Kings (Stormlight Archive #1)	Brandon Sanderson	Fantasy	4.65	249,376
Crooked Kingdom (Six of Crows #2)	Leigh Bardugo	Fantasy	4.61	137,335
Harry Potter and the Deathly Hallows	J.K. Rowling	Fantasy	4.61	2,530,201
Percy Jackson and the Olympians (1-3)	Rick Riordan	Fantasy	4.58	37,354
Clockwork Princess (Infernal Devices #3)	Cassandra Clare	Fantasy	4.58	298,801
The Nightingale	Kristin Hannah	Historical	4.57	558,801
Queen of Shadows (Throne of Glass #4)	Sarah J. Maas	Fantasy	4.57	193,374

INSIGHTS:

- Fantasy dominates the top-rated list, with 8 of the 10 highest-rated qualifying books belonging to this genre.
- Brandon Sanderson and Sarah J. Maas each place two titles among the top 10, reflecting strong reader reception for both authors' series.
- Harry Potter and the Deathly Hallows stands out for combining a very high rating (4.61) with an exceptionally large rater base (2.5 million), unlike most other top-10 entries which have far fewer raters.

15. Edition Language Distribution

The dataset records **11 distinct edition languages**, though it is overwhelmingly **English — 1,459 of 1,500 books (97.3%)**. A small number of records (21) show an "unknown" language value, and this analysis excludes one clearly mislabeled entry.

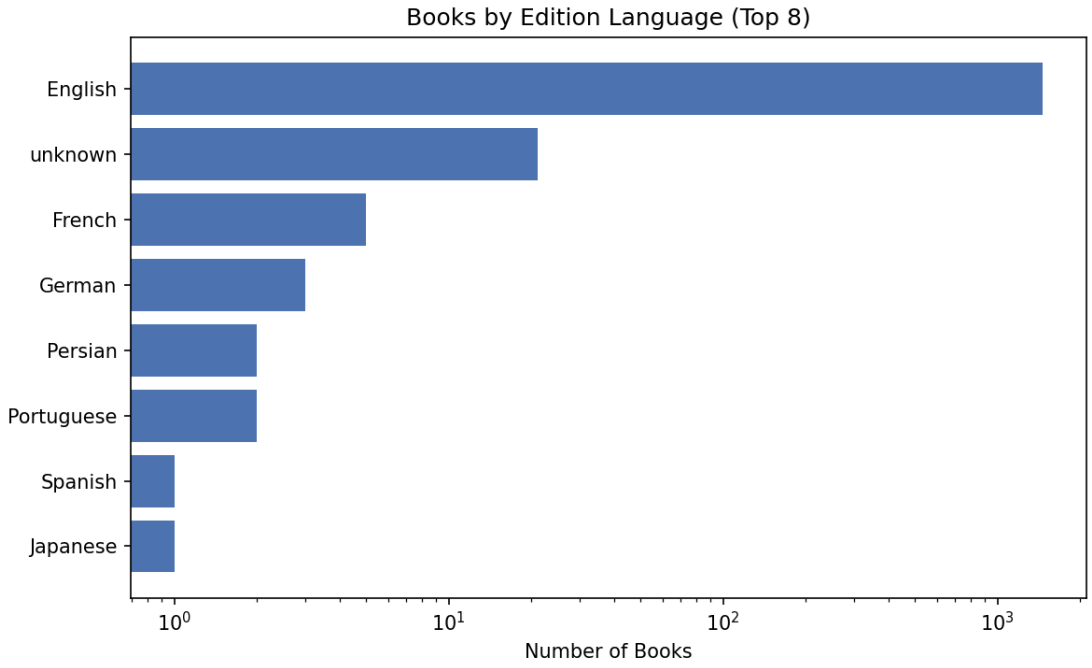


Figure 12. Books by edition language (top 8, log scale)

INSIGHTS:

- The near-total dominance of English editions reflects either the source platform's regional focus or how this particular "best books" list was curated.
- Non-English languages each contribute only a handful of records (5 or fewer), too few for reliable rating comparisons by language.
- This concentration means language-based segmentation is not a meaningful lens for this dataset without a broader, more internationally representative sample.

KPI :

Top language: **English**

Share of dataset: **97.3%**

Distinct languages recorded: **11**

16. Top Books by Goodreads List Score

The Goodreads **score** field reflects list-ranking performance rather than the raw star rating. **Harry Potter and the Deathly Hallows** tops the list with a score of **392,793**, ahead of The Hunger Games and The Kite Runner.

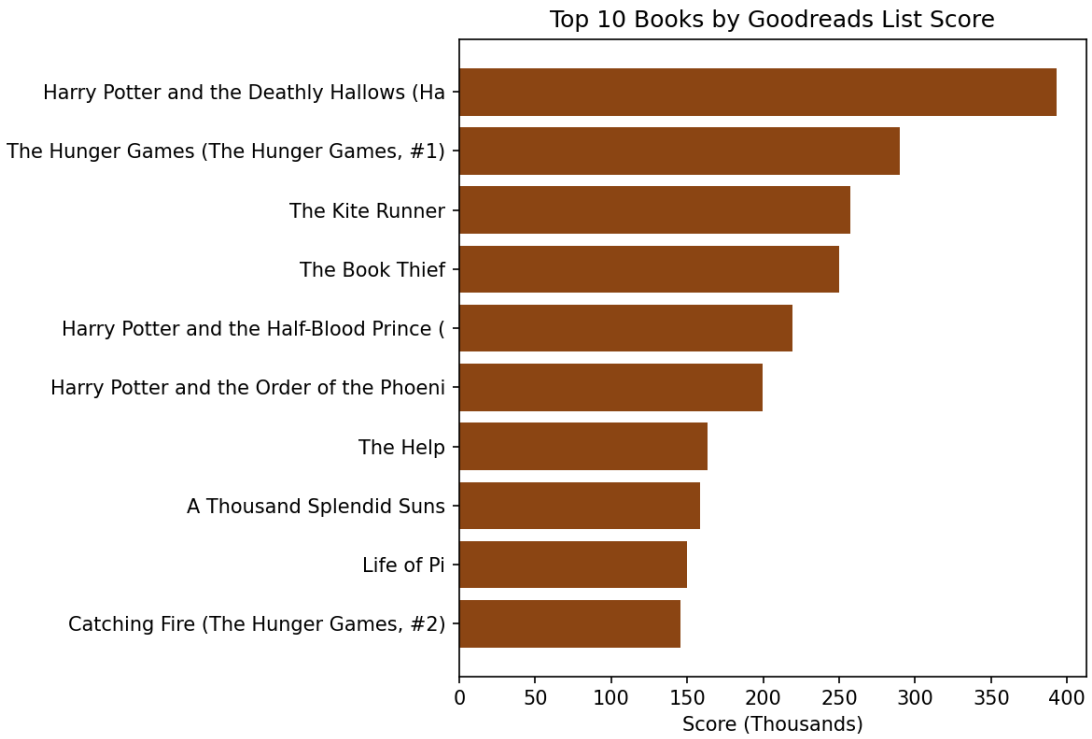


Figure 13. Top 10 books by Goodreads list score

INSIGHTS:

- Score and star rating measure different things: several of the top-scored books (e.g., The Kite Runner, The Book Thief) do not appear among the top-10 highest-rated titles, showing that popularity and average reader satisfaction do not always align.
- The Harry Potter series places three titles in the top 10 by score, reflecting sustained franchise-wide popularity beyond any single book.
- High-score books tend to be widely-read literary fiction and franchise titles rather than niche genre entries, unlike the ratings leaderboard which skewed heavily toward Fantasy.

KPI :

Top-score book: **Harry Potter and the Deathly Hallows**

Score: **392,793**

Author: **J.K. Rowling**

17. Author Rating Consistency (5+ Books)

Among authors with at least **5 books** in the list, **Sarah J. Maas** has the highest average rating at **4.46** across 8 books, closely followed by Brandon Sanderson at 4.45 across 7 books.

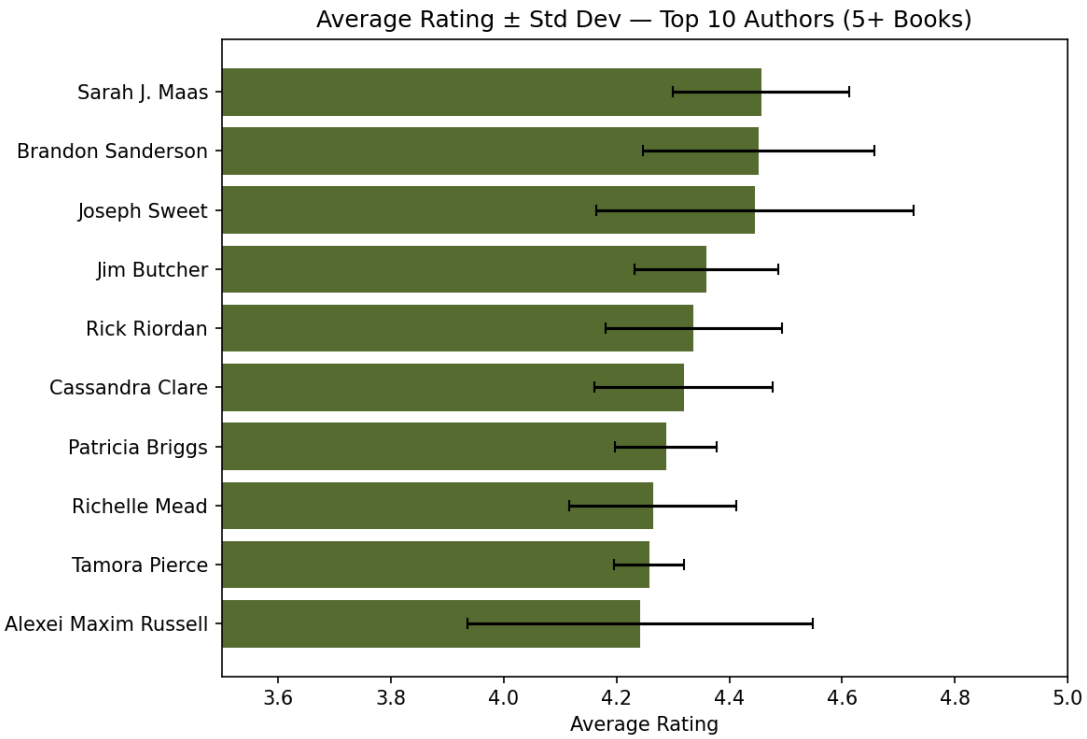


Figure 14. Average rating ± standard deviation — top 10 authors with 5+ books

INSIGHTS:

- Tamora Pierce shows the tightest consistency (std dev ≈0.06) among the top 10, meaning her books are rated very similarly to one another.
- Alexei Maxim Russell shows the widest spread (std dev ≈0.31), indicating more variable reader reception across their books in this list.
- High mean rating combined with low variance (e.g., Patricia Briggs, Tamora Pierce) suggests dependable reader satisfaction across an author's catalogue.
- This view complements the raw book-count ranking by showing quality consistency, not just volume of output.

KPI :

Most consistent top author: **Tamora Pierce (std ≈0.06)**

Highest average (5+ books): **Sarah J. Maas (4.46)**

18. Distribution of List Scores

Unlike average rating, the Goodreads **score** field is heavily skewed. The median score across all 1,500 books is **639.5**, far below the mean of **5,447** — a gap caused by a small number of extremely high-scoring blockbuster titles pulling the average upward.

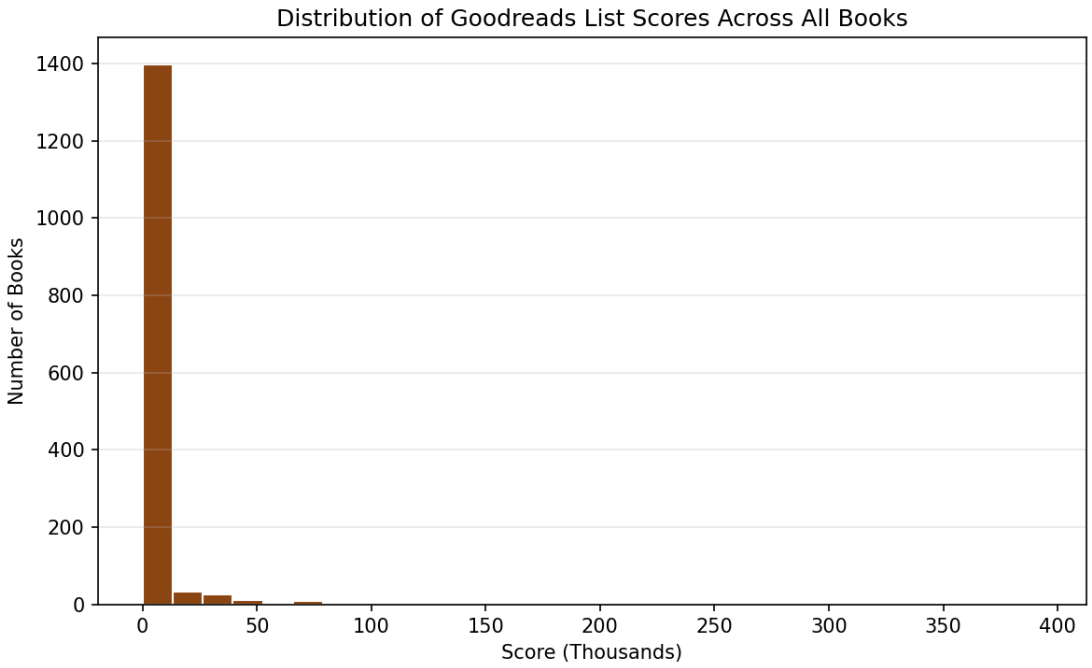


Figure 15. Distribution of Goodreads list scores across all books

INSIGHTS:

- Roughly 88% of books (1,313 of 1,500) have a score under 5,000, while the maximum single score (Harry Potter and the Deathly Hallows, 392,793) is nearly 80 times higher than that threshold.
- This right-skew means the mean score is not a representative "typical" value for this dataset — the median is a more reliable summary statistic here.
- The pattern is consistent with how Goodreads list-voting works: a handful of globally popular titles accumulate votes at a scale ordinary books never reach.
- Any future scoring-based ranking or forecasting work should treat score as a long-tailed variable rather than assume it is normally distributed.

KPI :

Median score: **639.5**

Mean score: **≈5,447**

Books scoring under 5,000: **1,313 (87.5%)**

Summary of Key Metrics

Metric	Value
Total books	1,500
Total genres	49
Total authors	873
Total raters (sum)	≈222.6 Million
Average rating	4.01 / 5
Median rating	4.02 / 5
Publication year range	2001 – 2020
Books with valid rating data	1,474 (98.3%)
Top genre by count	Fiction (385 books, 25.67%)
Top genre by engagement	Young Adult (≈58.5M raters)
Highest-rated genre (n≥20)	Fantasy (4.15 avg)
Most books by a single author	Rick Riordan / Stephen King (16 each)
Top-score book	Harry Potter and the Deathly Hallows (392,793)
Dominant edition language	English (97.3%)
Corrupted records (excluded from rating stats)	26

Next Steps to Move from Descriptive → Predictive

- Rating-trend forecasting by genre and author to estimate how future titles might be received.
- Anomaly detection for unusually low rater counts relative to book prominence or vice versa.
- Data-quality remediation pipeline to systematically catch and correct corrupted fields (e.g., the 26 rows with non-numeric rating values) before analysis.
- Author productivity and consistency analysis using rating variance across an author's full catalogue.
- Cohort analysis by decade or genre when a longer, more complete publishing history becomes available.

This report answers what the supplied Goodreads dataset shows. The next analytical layer would use a larger, more complete catalogue and richer metadata (page counts, series data, review text) to explain why rating and engagement patterns occur and to forecast reception for upcoming titles.

Glossary

I. Avg Rating — The mean of all individual reader ratings for a book on Goodreads, typically on a 1–5 scale.

II. No. of Raters — The total count of Goodreads users who have submitted a numeric rating for a given book.

III. Score — Goodreads' internal ranking score for a book on curated lists, generally derived from a combination of votes and ranking algorithm weighting rather than the raw star rating alone.

IV. No. of People Voted — The number of users who cast a list-ranking vote for the book (distinct from star ratings), used to compute the list-level score.

V. Genre — The primary category Goodreads assigns to a book based on user shelving and tagging behavior; a book may plausibly fit multiple genres, but this dataset records one primary genre per title.